SECTIONS OVER A FLAT FINITELY PRESENTED MORPHISM AS A HOMOTOPY COLIMIT OF PERFECT OBJECTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For a flat morphism of finite presentation the cohomology of the derived pullback is computed by the derived tensor of the pushforward of the structure sheaf, functorially in K. used at proof lines 8629-8633. Converts RGamma(U, Lf^*K) into RGamma(X, Rf_*O_U tensor^L K), the input to the colimit argument.
- The global sections functor on a quasi-compact separated scheme commutes with homotopy colimits because it commutes with direct sums, and tensoring with a fixed complex commutes with derived colimits because direct sums in the derived category are computed termwise. invoked at proof lines 8633-8641. Justifies passing the homotopy colimit through both sides of the desired identity.
- Morphisms of finite presentation and flatness descend along the approximation of the base ring by finitely generated subrings, and the derived pullback of the pushforward recovers the original pushforward under base change. invoked at proof lines 8647-8659. Reduces to the case where U and X are of finite type over Z.
- For a morphism of finite type over Z the pushforward of the structure sheaf is a homotopy colimit of perfect complexes as soon as the sheaves R^if_*O_U have countable sets of sections over affine opens, and the structure sheaf satisfies this countability. invoked at proof lines 8661-8670. Supplies the existence of the system of perfect objects E_n in the finite type case.

Source reference. Original lemma 08EU: sources/stacks-perfect/pinned-original-source.tex, source lines 5269–5269. Cited at proof lines 8629-8633 to rewrite RGamma(U, Lf^*K); the colimit reduction using it is inside the counted span.

Source reference. Original lemma 08DZ: sources/stacks-perfect/pinned-original-source.tex, source lines 803–803. Cited at proof lines 8635-8637 for RGamma(X, -) commuting with direct sums; the conclusion that it commutes with homotopy colimits is drawn inside the counted span.

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Source reference. Original lemma 08DQ: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 8643-8646 to obtain (1) from (2) by setting L_n = E_n^dual; the setting of the dual system is inside the counted span.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 8649-8653 to descend U to X to Spec(A_i); the use of the descended data is inside the counted span.

Source reference. Original lemma 04AI: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 8654-8656 to assume f_i flat; the reduction to f_i is inside the counted span.

Source reference. Original lemma 08IB: sources/stacks-perfect/pinned-original-source.tex, source lines 5362–5362. Cited at proof lines 8657-8659 to identify Rf_*O_U with the pullback of Rf_{i,*}O_{U_i}; the commutation of Lg^* with colimits and the resulting reduction are inside the counted span.

Source reference. Original lemma 0CRQ: sources/stacks-perfect/pinned-original-source.tex, source lines 8557–8557. Cited at proof lines 8662-8665 to reduce the existence of E_n to countability; the application and the countability conclusion are inside the counted span.

Source reference. Original lemma 0CRP: sources/stacks-perfect/pinned-original-source.tex, source lines 8508–8508. Cited at proof lines 8666-8670 applied to the structure sheaf; the final assembly of the argument is inside the counted span.

Lemma 0.2. *Let A be a ring. Let X be a scheme of finite presentation over A. Let $f : U \to X$ be a flat morphism of finite presentation. Then*

(1) *there exists an inverse system of perfect objects L_n of $D(\mathcal{O}_X)$ such that*
$$R\Gamma(U, Lf^*K) = \text{hocolim } R\mathrm{Hom}_X(L_n, K)$$
in $D(A)$ functorially in K in $D_{QCoh}(\mathcal{O}_X)$, and

(2) *there exists a system of perfect objects E_n of $D(\mathcal{O}_X)$ such that*
$$R\Gamma(U, Lf^*K) = \text{hocolim } R\Gamma(X, E_n \otimes^{\mathbf{L}} K)$$
in $D(A)$ functorially in K in $D_{QCoh}(\mathcal{O}_X)$.

Proof. By Lemma 08EU we have
$$R\Gamma(U, Lf^*K) = R\Gamma(X, Rf_*\mathcal{O}_U \otimes^{\mathbf{L}} K)$$
functorially in K. Observe that $R\Gamma(X, -)$ commutes with homotopy colimits because it commutes with direct sums by Lemma 08DZ. Similarly, $- \otimes^{\mathbf{L}} K$ commutes with derived colimits because $- \otimes^{\mathbf{L}} K$ commutes with direct sums (because direct sums in $D(\mathcal{O}_X)$ are given by direct sums of representing complexes). Hence to prove (2) it suffices to write $Rf_*\mathcal{O}_U = \text{hocolim} E_n$ for a system of perfect objects E_n of $D(\mathcal{O}_X)$. Once this is done we obtain (1) by setting $L_n = E_n^\vee$, see Cohomology, Lemma 08DQ.

Write $A = \operatorname{colim} A_i$ with A_i of finite type over $\mathbf{Z}$. By Limits, Lemma 01ZM we can find an i and morphisms $U_i \to X_i \to \operatorname{Spec}(A_i)$ of finite presentation whose base change to $\operatorname{Spec}(A)$ recovers $U \to X \to \operatorname{Spec}(A)$. After increasing i we may assume that $f_i : U_i \to X_i$ is flat, see Limits, Lemma 04AI. By Lemma 08IB the derived pullback of $Rf_{i,*}\mathcal{O}_{U_i}$ by $g : X \to X_i$ is equal to $Rf_*\mathcal{O}_U$. Since Lg^* commutes with

derived colimits, it suffices to prove what we want for f_i. Hence we may assume that U and X are of finite type over $\mathbf{Z}$.

Assume $f : U \to X$ is a morphism of schemes of finite type over $\mathbf{Z}$. To finish the proof we will show that $Rf_*\mathcal{O}_U$ is a homotopy colimit of perfect complexes. To see this we apply Lemma 0CRQ. Thus it suffices to show that $R^if_*\mathcal{O}_U$ has countable sets of sections over affine opens. This follows from Lemma 0CRP applied to the structure sheaf. □